

KiCad Implementation Guide

Custom UNO Breakout Carrier with RPi Camera Support

Project Overview

This guide walks through creating the schematic in KiCad for a custom breakout board based on the Arduino UNO Breakout Carrier (ASX00085) with added RPi camera support.

Part 1: KiCad Project Setup

1.1 Create New Project

1. Open KiCad 7.0 or later
2. File → New Project → New Project
3. Name: `UNO_Breakout_Custom`
4. Location: Choose your project directory

1.2 Configure Symbol Libraries

Add these libraries (Preferences → Manage Symbol Libraries):

- **Connector_FFC-FPC** (for J3 camera connector)
 - **Interface_Expansion** (for level shifters)
 - **Connector_Generic** (for headers J1, J2, J4, J5)
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Part 2: Schematic Symbol Selection

2.1 Main Connectors

J1 - JMEDIA (2x30 pins, 1.27mm pitch)

- Symbol: `Connector_Generic:Conn_02x30_Odd_Even`
- Footprint: `Connector_PinHeader_1.27mm:PinHeader_2x30_P1.27mm_Vertical`
- Alternative: Custom footprint for DF40C-60DP-0.4V(51) from Hirose

J2 - JMISC (2x30 pins, 1.27mm pitch)

- Same as J1

J3 - RPi Camera FFC (22-pin, 0.5mm pitch)

- Symbol: `Connector_FFC-FPC:Molex_52271-2279_1x22-1MP_P0.5mm_Horizontal`
- Footprint: `Connector_FFC-FPC:Molex_200528-0220_1x22-1MP_P0.50mm_Horizontal`

J4, J5 - Breakout Headers (2x20 pins, 2.54mm pitch)

- Symbol: `Connector_Generic:Conn_02x20_Odd_Even`
- Footprint: `Connector_PinHeader_2.54mm:PinHeader_2x20_P2.54mm_Vertical`

2.2 Active Components

U1 - Level Shifter

- Symbol: `Interface_Expansion:TXS0108EPW`
- Footprint: `Package_S0:TSSOP-20_4.4x6.5mm_P0.65mm`
- Manufacturer: Texas Instruments
- Part Number: TXS0108EPWR

2.3 Passive Components

Decoupling Capacitors (C1-C6)

- Symbol: `Device:C`
- Value: 0.1μF
- Footprint: `Capacitor_SMD:C_0603_1608Metric`
- Voltage Rating: 16V minimum

Bulk Capacitors (C7-C8)

- Symbol: `Device:C`
- Value: 10μF
- Footprint: `Capacitor_SMD:C_0805_2012Metric`
- Voltage Rating: 16V minimum

I2C Pull-up Resistors (R1-R4)

- Symbol: `Device:R`
- Value: 10kΩ
- Footprint: `Resistor_SMD:R_0603_1608Metric`

Part 3: Creating the Schematic

3.1 Sheet Organization

Create hierarchical sheets:

1. **Main Sheet** - Overview and power distribution
2. **Camera_Interface** - J1 → U1 → J3 connections
3. **Breakout_Headers** - J2 → J4, J5 connections

3.2 Main Sheet Layout

Power Distribution Section:

- Power input symbols (VIN, +5V, +3V3, +1V8)
- Ground symbol
- Bulk capacitors C7, C8

Hierarchical Sheet Blocks:

- Camera Interface (link to camera sheet)
- Breakout Headers (link to breakout sheet)

3.3 Camera Interface Sheet (Detailed)

Place Components:

1. J1 connector (left side of sheet)
2. U1 level shifter (center)
3. J3 camera connector (right side)
4. Decoupling caps C1-C4

Wire Connections:

J1 to U1 (1.8V Side):

```
J1 Pin 22 (CCI_I2C_SDA1) → U1 Pin 1 (A1)
J1 Pin 24 (CCI_I2C_SCL1) → U1 Pin 2 (A2)
J1 Pin 53 (CCI_I2C_SDA0) → U1 Pin 3 (A3)
J1 Pin 51 (CCI_I2C_SCL0) → U1 Pin 4 (A4)
J1 Pin 21 (+1V8)          → U1 Pin 20 (VCCA)
U1 Pin 10                  → GND
```

U1 to J3 (3.3V Side):

```
U1 Pin 19 (B1) → R1 (10k to 3.3V) → J3 Pin 2 (CAM_SDA)
U1 Pin 16 (B4) → R2 (10k to 3.3V) → J3 Pin 4 (CAM_SCL)
U1 Pin 11 (VCCB) → +3V3
U1 Pin 9 (OE) → +3V3
```

Direct Connections J1 to J3:

```
J1 Pin 16 (SOC_CAM_MCLK0) → J3 Pin 3 (CAM_GPIO/CLK)
J1 Pin 58 (+3V3)          → J3 Pin 21, 22 (3.3V)
J1 Pin 1,2 (GND)          → J3 Pin 1,5,9,13,17,20 (GND)
```

Add Power Decoupling:

```
C1: 0.1µF between +1V8 and GND (near U1 pin 20)
C2: 0.1µF between +3V3 and GND (near U1 pin 11)
C3: 0.1µF between +3V3 and GND (near J3)
C4: 0.1µF between +3V3 and GND (near R1/R2)
```

Add Labels:

- Label all nets clearly: I2C_SDA_1V8, I2C_SCL_1V8, I2C_SDA_3V3, I2C_SCL_3V3
- Add notes: "MIPI lanes NOT available - I2C only"

3.4 Breakout Headers Sheet

Simple Pass-Through Connections:

Create a table showing J2 pin to J4/J5 mapping:

```
For each pin on J2:  
  If pin 1-40: connect to corresponding pin on J4  
  If pin 41-60: connect to pin (n-40) on J5
```

Example:

```
J2 Pin 1 → J4 Pin 1  
J2 Pin 2 → J4 Pin 2  
...  
J2 Pin 40 → J4 Pin 40  
J2 Pin 41 → J5 Pin 1  
J2 Pin 42 → J5 Pin 2  
...  
J2 Pin 60 → J5 Pin 20
```

Implementation in KiCad:

1. Place J2 on the left
2. Place J4 and J5 on the right
3. Use buses for cleaner routing:
 - Create bus "JMISC[1..60]"
 - Split into "JMISC_A[1..40]" → J4
 - Split into "JMISC_B[41..60]" → J5

3.5 Power Distribution

Create Power Symbols:

1. VIN (input 7-24V)
2. +5V_USB
3. +3V3
4. +1V8
5. VBAT (+3.8V)
6. VCOIN (coin cell backup)
7. GND

Add **PWR_FLAG** symbols to each power net to avoid ERC warnings.

Power Decoupling Strategy:

```
Near J1: C5 (0.1µF on +3V3)  
Near J2: C6 (0.1µF on +3V3)  
Near U1: C7 (10µF on +3V3), C8 (10µF on +1V8)
```

Part 4: Electrical Rules Check (ERC)

4.1 Common ERC Warnings to Expect

Input Pin not driven:

- J3 pins 6-20 (MIPI lanes) will show as unconnected

- Mark as "Not Connected" or add NC flag
- Add text note explaining MIPI not available

Power pins not connected:

- Ensure all power input pins have PWR_FLAG
- Check all capacitor pins are connected to power/ground

4.2 Resolving Issues

1. Right-click unconnected pins → Properties → Electrical Type → "No Connect"
2. Add PWR_FLAG to VIN, +3V3, +1V8, +5V
3. Add text annotations for design notes

Part 5: Board Layout Considerations

5.1 Component Placement (Left to Right)

[J3]	[J1]	[U1]	[J2]	[J4]	[J5]
RPi	JMEDIA	Level	JMISC	Break	Break
Cam		Shift		out	out

5.2 Critical Routing Rules

I2C Traces:

- Keep I2C traces short (<50mm if possible)
- Route SDA and SCL parallel, same length
- Minimum 0.2mm trace width
- Add ground pour between traces

High-Speed Signals (if using PSSI):

- PSSI data lines should be length-matched ($\pm 5\text{mm}$)
- Route as differential pairs if possible
- Keep away from switching power supplies
- 50Ω controlled impedance preferred

Power Routing:

- VIN: 1mm width minimum
- +3V3: 0.8mm width minimum
- +1V8: 0.5mm width minimum
- Use copper pours for GND (top and bottom)

5.3 Decoupling Capacitor Placement

Place decoupling caps as close as possible to:

- U1 VCCA and VCCB pins (<5mm)

- J1 power pins
- J3 power pins

Via Guidelines:

- Use 0.3mm drill, 0.6mm pad for signal vias
 - Use 0.5mm drill, 0.8mm pad for power vias
 - Place multiple ground vias near decoupling caps
-

Part 6: Design Verification Checklist

6.1 Schematic Checklist

- ☐ All connector pins numbered correctly
- ☐ Level shifter A-side connected to 1.8V signals
- ☐ Level shifter B-side connected to 3.3V signals
- ☐ Pull-up resistors on I2C lines (3.3V side)
- ☐ All power decoupling capacitors placed
- ☐ Ground connections verified
- ☐ NC pins marked as such
- ☐ ERC runs with 0 errors
- ☐ Design notes added for MIPI limitation

6.2 Component Verification

- ☐ Connector pin counts match: J1(60), J2(60), J3(22), J4(40), J5(40)
- ☐ Connector pitch correct: 1.27mm for J1/J2, 0.5mm for J3, 2.54mm for J4/J5
- ☐ Level shifter supports bidirectional I2C
- ☐ Capacitor voltage ratings adequate
- ☐ Resistor power ratings adequate (1/10W for 0603)

6.3 PCB Layout Checklist

- ☐ 4-layer board recommended (signal-ground-power-signal)
 - ☐ Board dimensions match requirements
 - ☐ Mounting holes placed correctly
 - ☐ Minimum trace width: 0.15mm
 - ☐ Minimum clearance: 0.15mm
 - ☐ Silkscreen labels readable
 - ☐ Design rule check (DRC) passes
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Part 7: Bill of Materials (BOM)

Ref	Qty	Value	Package	Manufacturer	Part Number
J1	1	2x30 1.27mm	-	Hirose	DF40C-60DP-0.4V(51)
J2	1	2x30 1.27mm	-	Hirose	DF40C-60DP-0.4V(51)

Ref	Qty	Value	Package	Manufacturer	Part Number
J3	1	22-pin FFC	0.5mm	Molex	52271-2279
J4	1	2x20 2.54mm	-	Amphenol	10129381-940002BLF
J5	1	2x20 2.54mm	-	Amphenol	10129381-940002BLF
U1	1	TXS0108E	TSSOP-20	TI	TXS0108EPWR
C1-C6	6	0.1μF	0603	Murata	GRM188R71C104KA01D
C7-C8	2	10μF	0805	Murata	GRM21BR61C106KE15L
R1-R4	4	10kΩ	0603	Yageo	RC0603FR-0710KL

Part 8: Known Limitations & Workarounds

8.1 MIPI CSI-2 Not Available

Problem: Standard Raspberry Pi cameras use MIPI CSI-2 differential pairs which are not exposed on JMEDIA.

Workarounds:

1. Use JMISC PSSI interface for parallel cameras
2. Use a camera module with I2C control only (no video)
3. Design custom adapter for available signals
4. Check if UNO Q has alternate MIPI connections

8.2 Voltage Level Considerations

Problem: Camera modules expect 3.3V I2C, UNO Q provides 1.8V.

Solution: U1 level shifter handles this translation. Ensure:

- VCCA side always powered before VCCB
- Direction is bidirectional (built into TXS0108E)
- Pull-ups on 3.3V side only (not on 1.8V side)

8.3 Signal Integrity

Problem: 1.27mm pitch connectors and level shifters add parasitic capacitance.

Mitigation:

- Keep traces short
- Use ground pour
- Add series termination if needed (22-33Ω)
- Test I2C at lower speeds first (100kHz)

Part 9: Testing & Bring-Up

9.1 Initial Power-On Tests

1. Before inserting UNO Q:

- Check continuity: GND to GND on all connectors
- Check isolation: VIN not shorted to +3V3 or +1V8
- Check capacitor polarity (if using polarized caps)

2. With UNO Q inserted, no camera:

- Measure +1V8 at U1 pin 20
- Measure +3V3 at U1 pin 11
- Verify OE (pin 9) is high

3. With camera connected:

- Check I2C communication with i2cdetect
- Verify camera device appears at expected address
- Monitor MCLK signal with oscilloscope

9.2 I2C Testing

```
bash

# On UNO Q Linux console
i2cdetect -y 1 # Scan I2C bus 1 (adjust bus number)
# Should show camera device at 0x10, 0x36, or similar
```

9.3 Troubleshooting

Camera not detected:

- Check pull-up resistors
- Verify level shifter powered correctly
- Check cable orientation on J3
- Confirm camera power (3.3V at pins 21-22)

I2C communication errors:

- Reduce I2C clock speed
 - Add stronger pull-ups (4.7kΩ instead of 10kΩ)
 - Check for shorts or opens with multimeter
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Part 10: Files to Generate

10.1 KiCad Project Files

- `UNO_Breakout_Custom.kicad_pro` - Project file
- `UNO_Breakout_Custom.kicad_sch` - Main schematic
- `Camera_Interface.kicad_sch` - Camera hierarchical sheet

- `Breakout_Headers.kicad_sch` - Breakout hierarchical sheet
- `UNO_Breakout_Custom.kicad_pcb` - PCB layout

10.2 Manufacturing Files

- Gerber files (RS-274X format)
- Drill files (Excellon format)
- Bill of Materials (CSV)
- Assembly drawing (PDF)
- Pick-and-place file (CSV)

10.3 Documentation

- Schematic PDF (for review)
 - PCB fabrication notes
 - Assembly instructions
 - Test procedures
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Appendix A: Pin Mapping Tables

See the main connection document for complete pin-by-pin mappings.

Appendix B: References

1. Arduino UNO Breakout Carrier Datasheet (ASX00085)
2. TXS0108E Datasheet - Texas Instruments
3. Raspberry Pi Camera Module Documentation
4. KiCad Documentation: <https://docs.kicad.org/>

Appendix C: Alternative Level Shifters

If TXS0108E is unavailable:

- **TXB0108** - Compatible, slightly different characteristics
- **LSF0108** - Load switch based, lower capacitance
- **PCA9306** - Dedicated I2C level shifter (2 channels only)

Appendix D: Contact Information

For questions about this design, refer to:

- Arduino Forum: <https://forum.arduino.cc/>
 - Arduino UNO Q Documentation
 - This repository's issues page
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